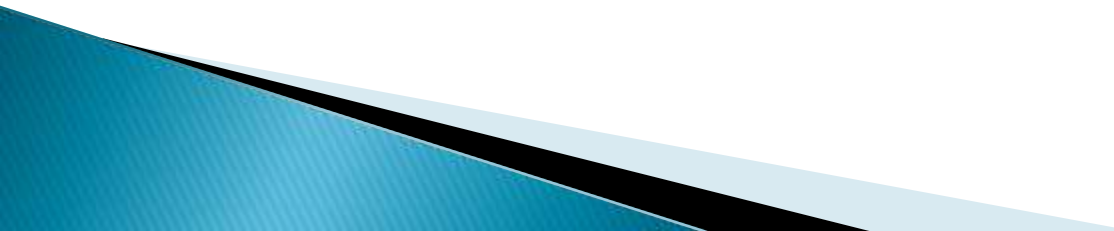


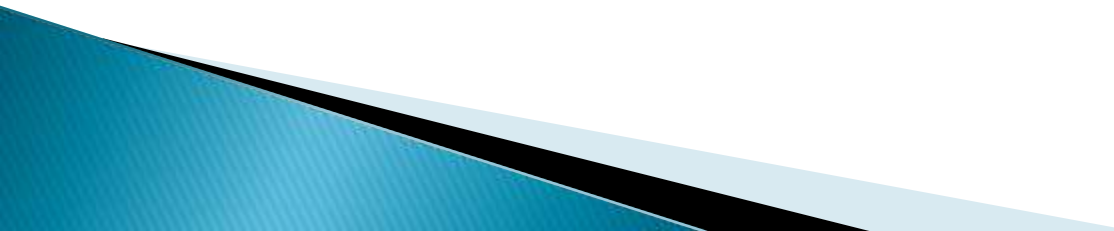
Mobile Application Testing

By
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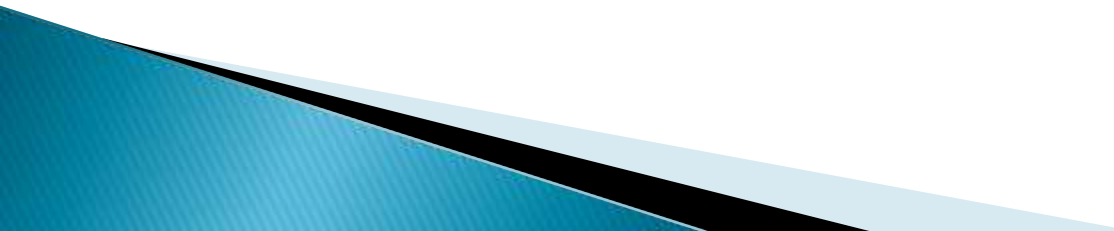
Topics

- ▶ What is Mobile Application Testing
 - ▶ Types of Mobile Application
 - ▶ Mobile Platforms
 - ▶ How to test Mobile Applications
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What is Mobile Testing

- ▶ **Mobile application testing** is a process by which application software developed for handheld mobile devices is tested for its functionality, usability and consistency.
 - ▶ It is similar to software testing but the testing will be performed on a mobile device instead of performing on a system.
 - ▶ Mobile applications either come pre-installed or can be installed from mobile software distribution platforms.
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Types of Mobile Application

- ▶ There are three types of Mobile Application:-
 - ▶ 1. Web Applications
 - ▶ 2. Native Applications
 - ▶ 3. Hybrid Applications
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What is Web Application

- ▶ Web apps are not real applications; they are actually websites that open in your smartphone with the help of a web browser.
- ▶ **Benefits:-**
 - ▶ 1. Easy Access
 - ▶ 2. No installation is required, as compared to native or hybrid app
- ▶ **Downside:-**
 - ▶ 1. Mobile websites cannot use some of the features. For example, access to the file system and local resources isn't available in websites.
 - ▶ 2. Users won't have the app's icon on their home screen as a constant reminder. The website needs to be opened in a web browser only.
 - ▶ 3. While native and hybrid apps appear on the App Store and Google Play, web apps won't. So redistribution is not that sensible.

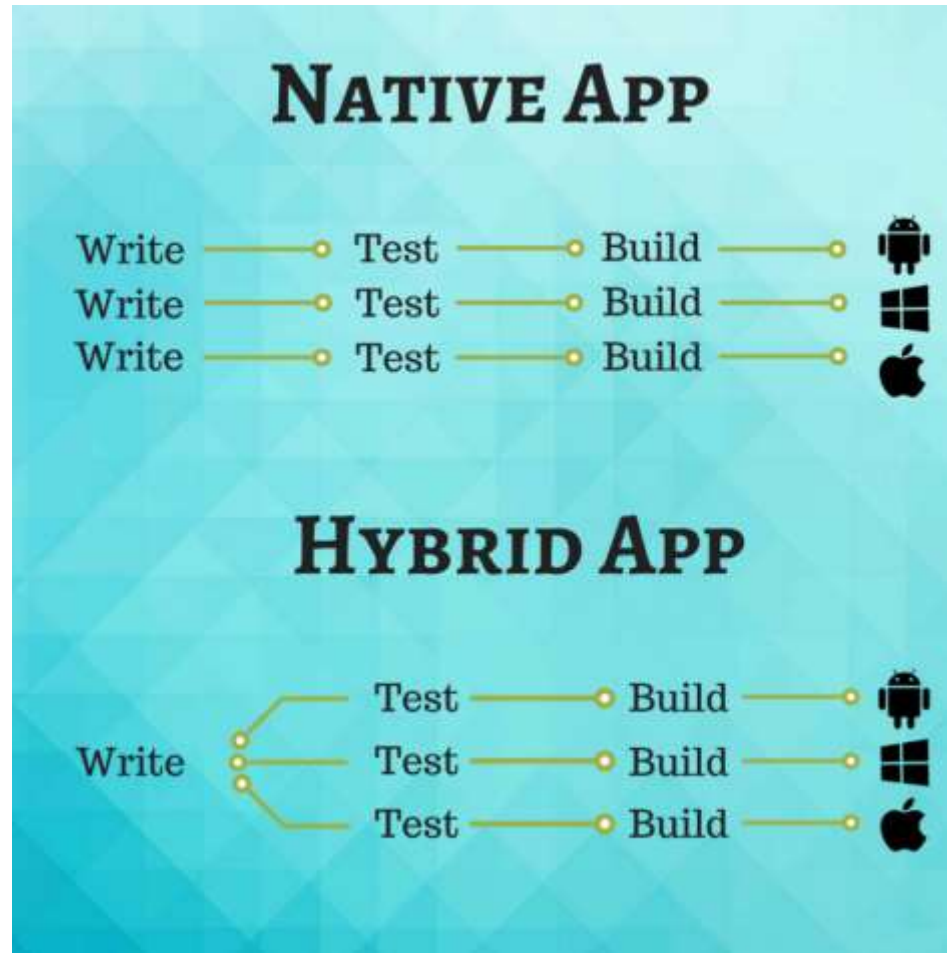
What is Native Application

- Native apps are developed for a specific mobile platform using particular programming languages and technologies. IOS apps, for example, are written in Objective-C and Swift, Android apps in Java or Kotlin. C++ is used for native Windows and Blackberry apps. **Example:- Instagram**
- This means that a separate version of the app must be developed for each platform. You can't reuse any piece of code from another platform version, as it is written in a completely different programming language. That's why native app development is considered to be the most time-consuming and most expensive.
- **Pros:-**
 - 1. Native mobile apps provide fast performance and a high degree of reliability.
 - 2. Users can use some apps without an internet connection.(Ex. Video games)
- **Cons:-**
 - 1. Expensive to develop.
 - 2. Long development time

What is Hybrid Application

- ▶ A hybrid app is a program that is built using HTML 5, CSS and JavaScript and wrapped in native container. The native container loads maximum information on the page as soon as the user navigates the application. Example:-LinkedIn
- ▶ If a native app and a web app got married and had a kid, it would be a hybrid app.
- ▶ **Advantages:-**
 - ▶ 1. Reusing the code:- the code is written once and deployed across all mobile platforms.
 - ▶ 2. Reducing development time and cost:- the code is written once, which substantially reduces development time and costs compared to native apps which require development for iOS and development for Android.
- ▶ **Disadvantage:-**
 - ▶ 1. Slower performance and transition between pages.
 - ▶ 2. Dependency on the browser's speed.

Native App vs Hybrid App



Key differences between Native and Hybrid Apps

Features	Native Apps	Hybrid Apps
Cross Platform Support	No	Yes
Speed	Very Fast	Good
Security	High	Low
Graphics	Better Graphics	Lesser graphics than native apps
Internet Connection	Not required always	Required
Integration	The camera, address book, geolocation, and other features native to the device can be seamlessly integrated into native apps.	Some device features may be harder to integrate into hybrid apps.

Types of Applications in Mobile

- ▶ **Types of Applications in Mobile:-**

- ▶ 1. Pre-installed Applications
- ▶ 2. Installable Applications

- ▶ **1. Pre-installed Applications:-**

- ▶ Applications which are shipped as in built software with the mobile device
- ▶ No installation/Un-installation involved
- ▶ Crashes can cause severe damage to ROM.

- ▶ **2 . Installable Applications:-**

- ▶ Can be upgraded
- ▶ Can be installed/Un-installed from the device
- ▶ Can be downloaded from App Store or OTA(Over the Air) the URL which leads to download the application.

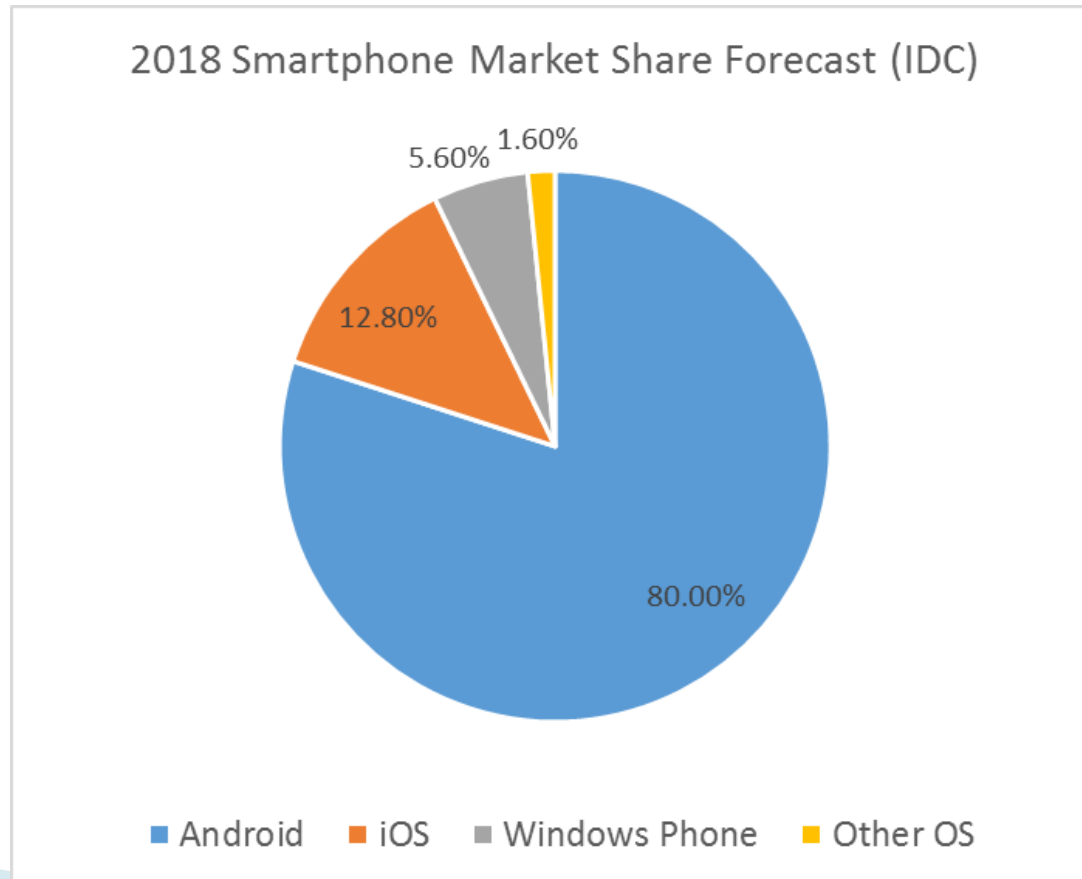
Mobile Operating Systems

- ▶ The following table gives an overview of some of the popular mobile operating systems available in market:

Operating System	Developed by	Popularity (Low, Medium, High)	Latest available version
Android	Google Inc	High	9 (Pie)
IOS	Apple Inc	High	11.4
Blackberry	Blackberry ltd	Low	Blackberry 10.3.3
Windows	Microsoft Inc	Medium	Windows 10
Symbian	Symbian foundation	Low	Discontinued

Mobile Operating System Market Share

- ▶ Based on some generic survey, the usage of different operating systems in the market can be depicted as shown below.



Android vs IOS

- ▶ **1. Device Choice** :- With Android we have much greater choice of handsets for consumers in terms of device and screen size, price and colour.

- ▶ **2. The home screen on iOS is not as customizable as the one on Android:-**
 - ▶ 2.1 On my iPhone, the only things I can change about the home screen are:
 - ▶ 2.1.1 Choosing a different image for the wallpaper
 - ▶ 2.1.2 Changing the order of the app icons
 - ▶ 2.1.3 Creating folders in which I can place multiple app icons

 - ▶ 2.2 That is not much compared to what you can do in Android. These are the features I miss most:
 - ▶ I cannot move app icons around and place them anywhere I want on the home screen
 - ▶ I cannot use widgets. The only place where you can have widgets in iOS is the Notifications Screen.

Android vs IOS

▶ 3. Design:-

- ▶ iPhone does not have back button.
- ▶ The title in Android applications is normally placed at the top left of the screen, whereas iOS app's title will be centered.
- ▶ Android uses a drawer menu as a navigation pattern, whereas iOS' pattern is a tab bar.

▶ 4. Media files cannot be transferred to a Windows PC, as easily, when using an iPhone:-

- ▶ If you are a Mac user, transferring your files is not an issue. However, if you are a Windows PC user, moving media files on and from an iPhone can be daunting at first. With an Android device, all you have to do is plug it into a USB port on your Windows computer. Things are different with iPhones. If you want to transfer music or video files, you must download and install iTunes and use this app instead of File Explorer. Also, when I wanted to copy some photos from my iPhone to my Windows 10 PC, I had to search for a way to do it. So far, the best method I found for doing that is to use the [Microsoft Photos](#) app to import pictures from the iPhone.

Mobile Device Types

- ▶ Take a look at the following table. It differentiates tablets, e-book readers, and smartphones based on their characteristics.

Device	Tablets	E-book Readers	Smart Phones
What it is	Tablets are portable computer devices. Unlike traditional computers, they don't have keyboards or mouse, however the entire screen is touch sensitive.	E-book readers—also called e-readers—are similar to tablet computers, except they are mainly designed for reading e-books (digital, downloadable books).	A smartphone is a powerful mobile phone that is designed to run a variety of applications in addition to providing phone service.
Used for	Almost all the jobs which we can do with traditional computers or desktops.	Reading e-books	Web browsing, watching videos, reading e-books, and playing games
Example	Samsung Tablets	Amazon Kindle, Barnes & Noble Nook.	Sony smartphones, Samsung smartphones, Apple iPhone.

How to test a Mobile Application

How to test a Mobile Application ?

Installation: Verify that application can be Installed Successfully.

Types of installation

1. OTA(Over The Air)– Need GPRS sim card and internet connection.
2. Wi-Fi
3. Data cable
4. Bluetooth

Uninstallation: Verify that application can be uninstalled successfully.

Application Logo:

Verify that application logo with Application Name is present in application manager and user can select it.

Splash:

Verify that when user selects application logo in application manager splash is displayed.

Note: Splash do not remain for more than 3 seconds.

How to test a Mobile Application

- ▶ **Functional Testing:-**

- ▶ Functional testing ensures that the application is working as per the requirements.

- ▶ **Memory Leakage Testing:-**

- ▶ A memory leakage is the type of resource leak that occurs when a computer program incorrectly manages memory allocations in such a way that memory which is no longer needed is not released. A memory leak reduces the performance of the computer by reducing the amount of available memory.

How to test a Mobile Application

- ▶ Memory Leakage Example:-
 - ▶ When a button is pressed:
 - ▶ Get some memory, which will be used to remember the floor number Put the floor number into the memory
 - ▶ Are we already on the target floor?
 - ▶ If so, we have nothing to do: finished
 - ▶ Otherwise:
 - ▶ Wait until the lift is idle
 - ▶ Go to the required floor
 - ▶ Release the memory we used to remember the floor number
 - ▶ The memory leak would occur if the floor number requested is the same floor that the elevator is on; the condition for releasing the memory would be skipped. Each time this case occurs, more memory is leaked.
 - ▶ The memory leak lasts until the system is reset. For example: if the elevator's power were turned off or in a power outage, the program would stop running. When power was turned on again, the program would restart and all the memory would be available again, but the slow process of memory leak would restart together with the program

How to test a Mobile Application

- ▶ The leak in the above example can be corrected by bringing the 'release' operation outside of the conditional:
- ▶ When a button is pressed:
 - ▶ Get some memory, which will be used to remember the floor number
 - ▶ Put the floor number into the memory
 - ▶ Are we already on the target floor?
 - ▶ If not:
 - ▶ Wait until the lift is idle
 - ▶ Go to the required floor
 - ▶ Release the memory we used to remember the floor number
 - ▶

How to test a Mobile Application

- ▶ **Updates Testing:-**

- ▶ People frequently complain about applications not working satisfactorily after an update. So it is very important that under the update testing, we qualify that the App will work as it was working previously.

- ▶ **Interrupt Testing:-**

- ▶ An application, while functioning, may face several interruptions like incoming calls or network coverage outage and recovery. This can again be distinguished for:
 - ▶ Incoming and Outgoing SMS and MMS
 - ▶ Incoming and Outgoing calls
 - ▶ Battery Removal
 - ▶ Cable Insertion and Removal for data transfer

How to test a Mobile Application

- ▶ **Power Consumption Testing:-**

- ▶ While we focus on power consumption testing, we are required to measure the state of the battery at each activity level. It will give us a better understanding of power consumption by an individual application. Power Consumption test can be done manually; also there are some free tools available in the market such as Trepan Profiler, Power Tutor, and Nokia Energy Profiler. These are applications which can display the real-time power consumption on a smartphone or tablet.
- ▶ Example:- Keep the mobile app in ideal condition; verify that there is no power consumption when there is no activity happening for the app.

- ▶ **Security Testing:-**

- ▶ In this type of testing, we verify that mobile app must be secure. **Example:-** Applications should encrypt user name and passwords when authenticating the user over a network.
- ▶ Verify that the application does not get operated with same user credentials on two different mobile devices.
- ▶ Verify that a session automatically gets expired if it remains inactive for more than

How to test a Mobile Application

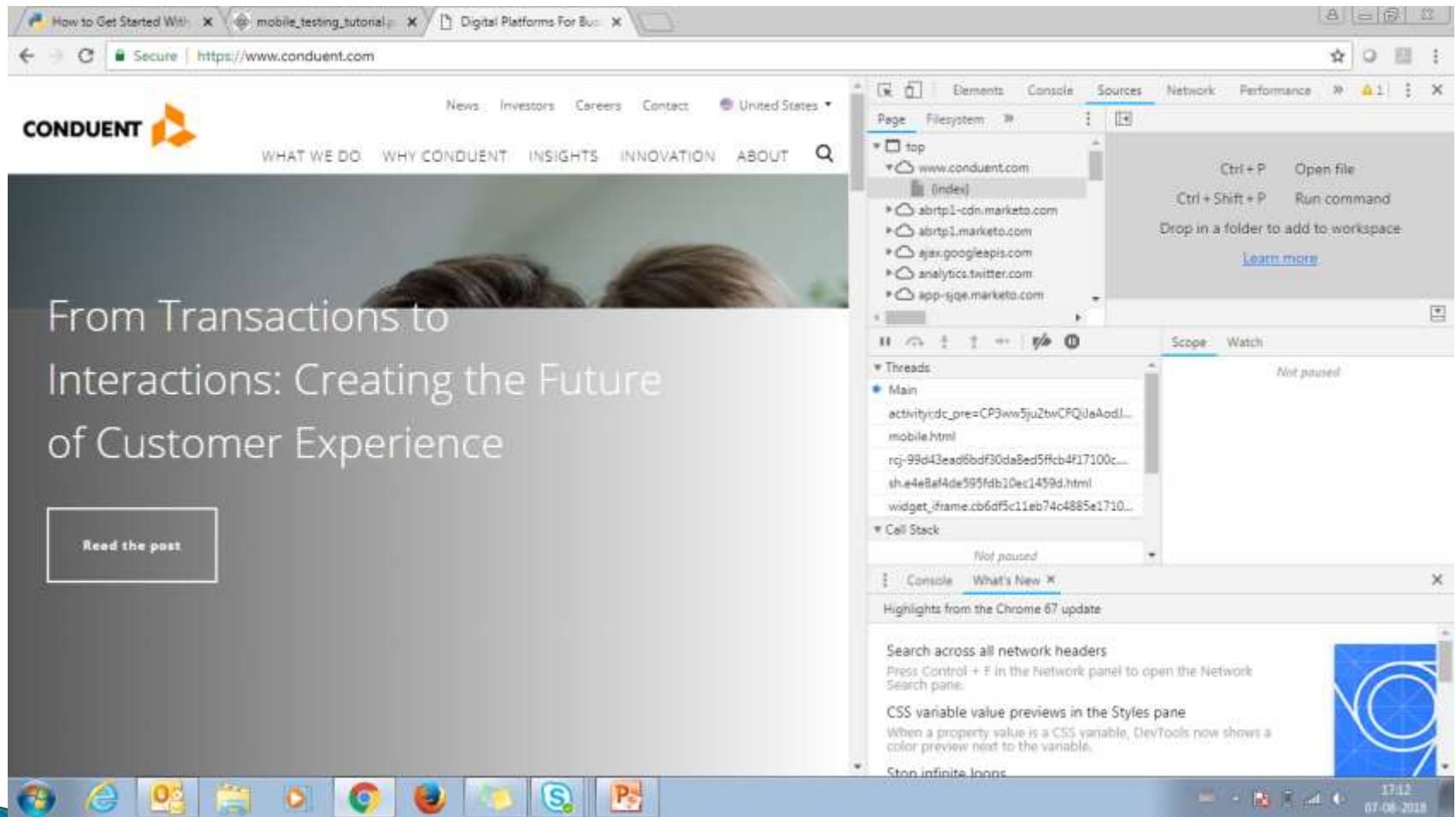
- ▶ **Screen Orientation/resolution:–**
- ▶ There are different screen sizes for mobile applications so it is difficult to test an application against resolution. Standard screen resolutions are given below:-
 - ▶ 640 × 480
 - ▶ 800 × 600
 - ▶ 1024 × 768
 - ▶ 1280 × 800
 - ▶ 1366 × 768
 - ▶ 1400 × 900
 - ▶ 1680 × 1050

How to test a Mobile Application

- ▶ **Available Tools:** –

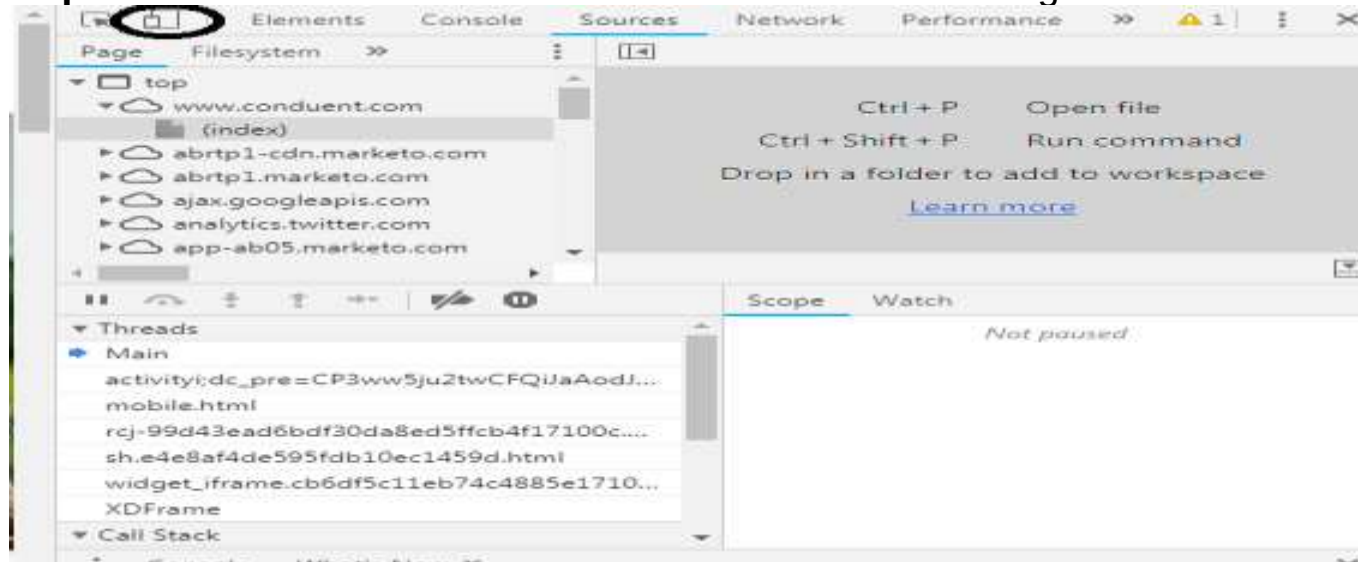
- ▶ There are quite a few tools available in the market to make mobile UI testing smoother and simpler. For example:
 - ▶ Google chrome extension
 - ▶ Screen fly
 - ▶ Browser Stack
- ▶ Here we will talk about only google chrome:-
 - ▶ **Step 1:** Open the website under testing in “Google Chrome Web browser”.
 - ▶ **Step 2:** Press F12. It will open the Developer tool window, as shown in the following screenshot

How to test a Mobile Application



How to test a Mobile Application

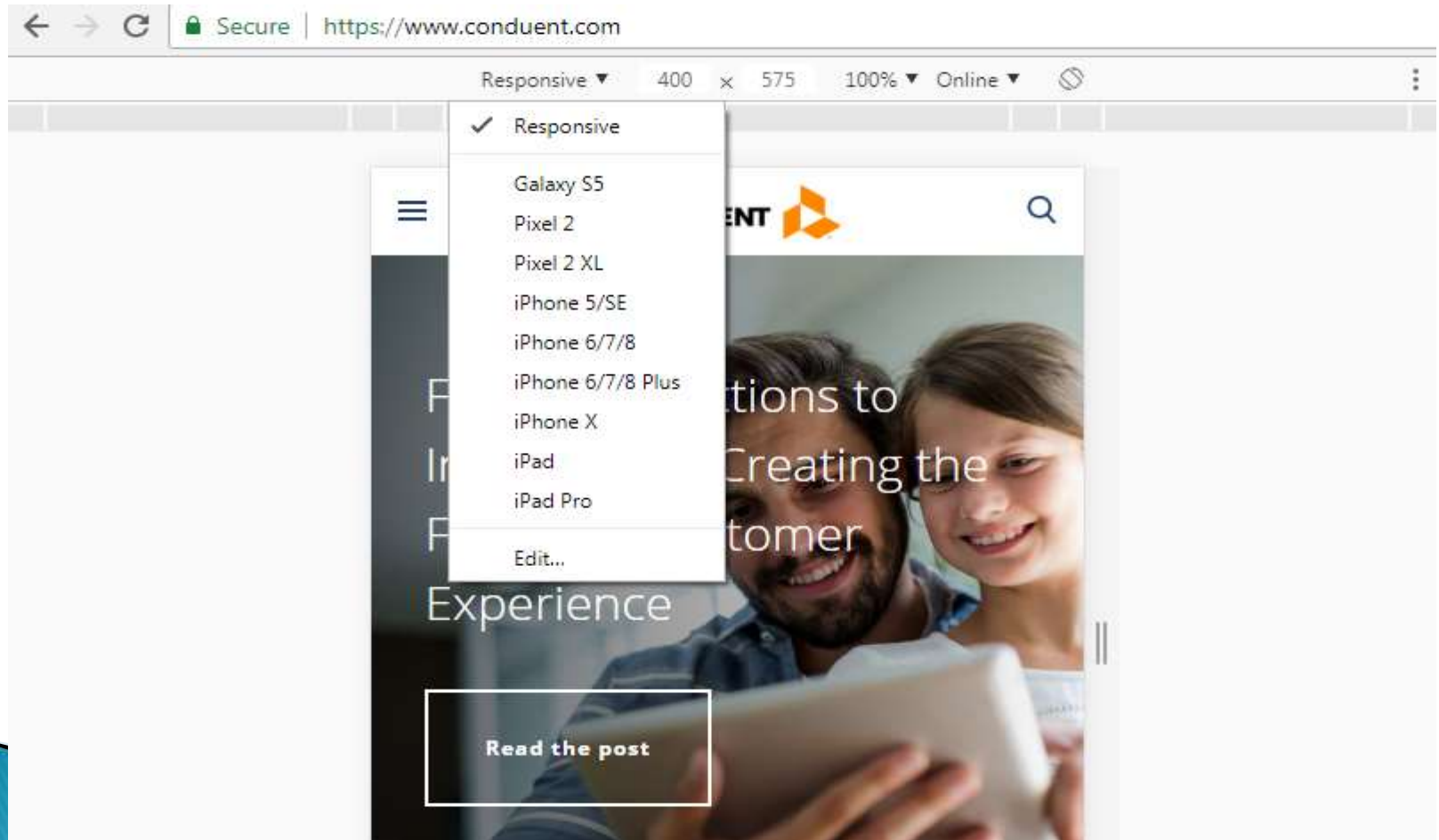
- ▶ **Step 3:** Click the mobile device like icon. Refer the following screenshot.



- ▶ **Note:-** Just open the site and press Ctrl+Shift+M to see the mobile view.

How to test a Mobile Application

- ▶ **Step 4:** Select the mobile device with which you want to test the website. You can choose the different available devices in order to do the UI verification.



Some of the points to consider while doing Mobile Testing

Some of points to consider while doing Mobile Testing:

1. Testing in Network strengths like Low, Medium, High
2. Testing in Change of Network strength from low to high and high to low
3. Testing in various Battery Strengths like Critical, Low , High and During charging
4. Monitoring Battery Consumption pattern while running the application in foreground, Background and for long time
5. Interruptions: Activities which can occur parallel with in the device while application is being installed, launched, run, exit, uninstalled.
6. Examples of Interruptions like Receiving Incoming call, Lose Network connectivity and regain it, Activated Camera, Remove Battery, Shutdown the device.
7. Testing with different input modes in different platforms

Mobile Emulator

- ▶ Suppose we are developing an application for flight ticket booking system. Once the product is entirely developed, as a part of mobile testing, we need to check if the application is working as expected with all the majorly used devices like Android phones, iOS, Blackberry phones, and other different types of tablets and iPads.
- ▶ It would be difficult for us to purchase such a large number of mobile devices and carry out testing. So is there any smart alternate available?
- ▶ The solution to this problem is to use Mobile Simulators and Mobile Emulators.
- ▶ **A simulator/emulator cannot mimic the following features:**
 - ▶ 1. Mobile Device Battery
 - ▶ 2. Mobile Device's camera
 - ▶ 3. Difficult to mimic interruption like incoming calls and SMS.
- ▶ The Android Emulator tests both the hardware and the software whereas the Android Simulator tests only the software. It means you can test the associated parts of the product as well by using an Emulator. But it is not possible with a simulator.

Thank You

*Thank
you*

